Qatar-1b

R-1626

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_250525 · created 2026-08-02T05:06:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460820.7787 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.054
Transit depth (Rp/Rs)^2	2.9 +/- 1.85 [%]
Orbital Inclination (inc)	79.16 +/- 2.86
Airmass coefficient 1 (a1)	1.98 +/- 0.41
Airmass coefficient 2 (a2)	-0.37 +/- 0.17
Scatter in the residuals of the lightcurve fit is	21.55 %
Best Comparison Star	#1 - [323, 424]
Optimal Aperture	1.81
Optimal Annulus	14.09
Transit Duration (day)	0.024 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.17 ± 0.054 vs 0.1463 (deviation 0.25σ)
Duration fit vs published	0.024 d vs 0.0692 d (deviation 0.94σ)
Mid-time O–C	-4.9 min
Depth SNR	1.8
Residual scatter	21.55 %

Light curve

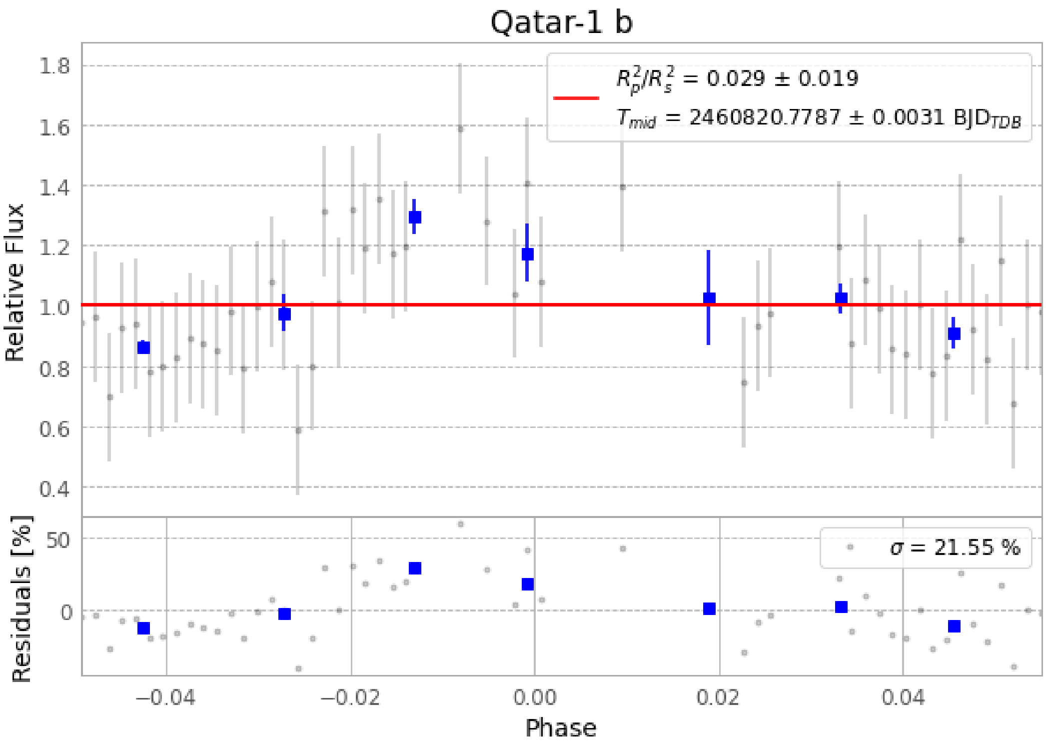

Figure 1 — FinalLightCurve_Qatar-1 b_2025-05-24.png (SHA-256 ec1e261e71057a49…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [362, 172], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 82257b8e68208ef7…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

72 FITS frames (47 MB), `Qatar-1250525050015.FITS` → `Qatar-1250525083315.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-250525.log (d57a593cfc2b…), FinalLightCurve_Qatar-1 b_2025-05-24.png (ec1e261e7105…), FinalLightCurve_Qatar-1 b_2025-05-24.csv (ef9732335b02…)

Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-02 05:06Z.